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PROFESSIONAL SUMMARY

A Senior Data Scientist (Senior ML Engineer) having total experience of 9+ years (6 years in ML/DL/NLP/GenAI) with knowledge of Python, SQL, Excel, Applied Statistics, Data Visualization, Data Wrangling, Predictive Modeling, Time Series Analysis, Machine Learning, NLP, Anomaly Detection, , LLM, Computer Vision, Generative AI across various domain like finance, healthcare, oil & gas, and telecom. Expert in end-to-end ML pipelines, production-grade GenAI/RAG systems, and secure, explainable AI solutions that deliver measurable business impact.

TECHNICAL SKILLS

Programming Languages: Python, SQL

Data Science Packages: Machine Learning, Statistics, EDA, Python Libraries, NLP, Text Analytics, Computer Vision, Generative AI

GenAI Tools: LLMs, ChatGPT, LLMs, OpenAI, Gemini, Groq API, Prompt Engineering, Ollama, Hugging Face,RAG

Analytics / Cloud Tools: AWS

Python, ML, CV & NLP Libraries: Pandas, NumPy, Scikit-Learn, Gensim, SpaCy, NLTK, BERT, Keras,Matplotlib, Seaborn etc.

WORK EXPERIENCE

Relevant experience: 6+ years

AI/ML developer (Senior Data Scientist) | Fractal.AI (Jul 2024 – Present)

Job Description:

Inhouse Project

- Spearheaded the design of an enterprise RAG pipeline using GPT-4.o Chroma DB, and LangChain, optimizing chunking strategies, semantic embeddings, and hybrid retrieval to improve internal document retrieval precision by ~20% and reduce hallucination rates by ~55%.
- Built FastAPI backend + Streamlit UI delivering explainable LLM outputs with citations & relevance scores which help in manual document search.
- RAG-based Chatbot: Developed a Retrieval-Augmented Generation chatbot with citation links redirecting to relevant PDF pages for improved transparency and accuracy.
- PDF-to-Podcast Generator: Built an AI pipeline for text summarization, transcript creation, audio narration, and video generation.

Client 1 (A reputed Core Industry)

- Handled large datasets, data coming from IOT sensors placed on high pressing machines, recorded at nanoseconds for various metrics and converted the data into usable form.

- As a part of EDA, outliers and missing values were handled.
- Aggregated the data at different time intervals and timestamp to check the distribution at different intervals.
- Designed an anomaly-detection pipeline for oil & gas equipment using **Convolutional Autoencoders** trained exclusively on **normal sensor telemetry** (temp, pressure, vibration, speed).
- Built a multi-level **asset hierarchy mapping (facility → machine → train → subsystem)** with SME-driven sensor selection for each model.
- Engineered a full **data ingestion workflow** using pandas from internal sensor tables, performed data masking using shutdown indicators, and generated reconstruction-error reports.
- Developed and validated system-specific models registered them in the **internal model registry**, and packaged artifacts for production deployment.
- Integrated models with live sensor streams to enable **real-time anomaly scoring**, automated alert generation, and backend storage powering the monitoring UI.

AI/ML developer (Senior Data Scientist) **Ernst & Young (EY) (Oct 2022 – Jul 2024)**

Job Description:

- Delivered GenAI solutions (GPT-3.5, GPT-4, BERT) to structure unstructured financial data for Fortune 50 clients → automated insights and query handling. Engaged in a cutting-edge Gen AI project focused on leveraging advanced algorithms to structure unstructured financial data from multiple companies. Tasked with transforming disparate data sources into a centralized structured database, facilitating streamlined access to essential insights and enabling efficient responses to external queries using gpt-3.5 and gpt-4.
- Conducted in-depth analysis of Fortune 50 companies' 10-K reports, focusing on risk identification such as Cyber security and human capital. Utilized NLP techniques for zero-shot classification to categorize risks into custom categories.
- Employed deep learning models i.e. BERT Topic to identify and analyse various topics present in 10-Kreports, providing a comprehensive understanding of the content.
- Developed an automated pipeline for client survey analytics using REST API, enabling seamless data retrieval and performed sentiment analysis on employee feedback using advanced BERT models and text analytics.
- Contributed to healthcare projects, including patient journey mapping, anomaly detection and drug discovery. Leveraged Random Forest to prioritize the top 10 drugs for laboratory testing, optimizing the drug discovery process.
- Utilized predictive modelling and machine learning (supervised learning) to achieve approximately 82% accuracy in predicting persistent and non-persistent patient categories. Incorporated multi class classification models and optimized parameters using grid search CV.
- Created a visualization framework using matplotlib, Plotly to analyse patent status (alive, granted, and inactive) and other quantitative metrics. Employed various charts like Tree Map, Sunburst, and stacked bar charts for comprehensive insights.
- Explored and identified innovation landscapes across multiple technologies in global markets. Applied NLP techniques and text analytics to analyse innovation density within each technology space, supporting strategic decision-making i.e. sentiment Analysis, NPS etc.
- Worked on data visualization using Python and Power BI, incorporating DAX and M queries to provide in-depth insights for clients. The refined visualizations enhance data interpretation and aid in informed decision-making.

Data Scientist | Concentrix Pvt. Ltd. (Jan 2020 – Oct 2022).

Client: Biggest Telecom Giant in USA

Job Description:

- Telecom Plan Recommendation System: Hybrid collaborative + content-based filtering with LightGBM learning-to-rank → 0.71 Precision@3, simulated 7% ARPU uplift.
- Built interactive dashboards and predictive analysis for credit collections, worker fatigue, and oil & gas productivity.
- Applied a variety of machine learning algorithms, including logistic regression, random forest, KNN, SVM and decision trees, to enhance text analytics capabilities.
- Conducted comprehensive data analysis, incorporating feature selection, dimensionality reduction technique, feature engineering, NLP techniques, dimensionality reduction, text analysis and statistical analysis.
- Developed and delivered interactive dashboards and reports, providing clients with actionable insights and recommendations based on project outcomes.
- Led initiatives to enhance credit card collections through various Proof of Concepts (POCs), leveraging data analysis and trend analysis to drive improvements.
- Implemented a Logistic Regression model for detecting worker fatigue within the Oil and Gas Industry, contributing to enhanced safety measures.
- Text Classification of data using NLTK, Machine Learning libraries, TF-IDF vectorizer, finetuning with transformer models.
- Data cleaning, stop word removal, lemmatization, tokenization using NLTK and other machine learning libraries. Pandas have been used for EDA and feature engineering.
- Sentiment analysis has been done by using ML algorithms by classifying text data.
- Use of data augmentation techniques and oversampling techniques on text data.
- Extraction of powerful words from text using TFIDF score.

Assistant Manager | IndusInd Bank Ltd. (June 2017 – Aug 2018)

Assistant Manager | Kotak Mahindra Bank Ltd. (Nov 2015 – May 2017**)**

• Managed customer portfolios handled CASA operations and resolved fraud-related cases efficiently.

Certifications

- Data Science and Machine Learning – Corpnce Pvt. Ltd.
- Advanced Power BI – Coursera

Education

Highest education: B Tech in electronic and telecommunication
Year of passing: 22/07/2014

PERSONAL INFORMATION

Date of Birth	: 10 January 1993
Language	: English, Hindi, Odia (Read, Write and Speak)
Gender	: Female
Nationality	: Indian